Week 2

Here are some matrices:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 7 \end{pmatrix}, B = \begin{pmatrix} 3 & -1 & 2 \\ -3 & 2 & 1 \end{pmatrix},$$
$$C = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, D = \begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix}, E = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

Find if possible $-3A, 3B - A, AC, CB, AE, EA$. If it is not possible explain why.

$$-3A = -3\underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 7 \end{bmatrix}}_{A} = \begin{bmatrix} -3\cdot 1 & -3\cdot 2 & -3\cdot 3 \\ -3\cdot 2 & -3\cdot 1 & -3\cdot 7 \end{bmatrix} = \begin{bmatrix} -3 & -6 & -9 \\ -6 & -3 & -21 \end{bmatrix}$$

$$3B - A = 3\underbrace{\begin{bmatrix} 3 & -1 & 2 \\ -3 & 2 & 1 \end{bmatrix}}_{B} - \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 9 & -3 & 6 \\ -9 & 6 & 3 \end{bmatrix}}_{3B} - \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 7 \end{bmatrix}}_{A} = \begin{bmatrix} 8 & -5 & 3 \\ -11 & 5 & -4 \end{bmatrix}$$

$$AC = \underbrace{\begin{bmatrix} \; & \; \end{bmatrix}}_{2\times 3}\underbrace{\begin{bmatrix} \; & \; \end{bmatrix}}_{2\times 2}$$

$\neq$

: Not possible because number of rows of C is not equal to the number of columns of A.

$$CB = \underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}}_{2\times 2}\underbrace{\begin{bmatrix} 3 & -1 & 2 \\ -3 & 2 & 1 \end{bmatrix}}_{2\times 3} = \underbrace{\begin{bmatrix} 3-6 & -1+4 & 2+2 \\ 9-3 & -3+2 & 6+1 \end{bmatrix}}_{2\times 3} = \begin{bmatrix} -3 & 3 & 4 \\ 6 & -1 & 7 \end{bmatrix}$$

$=$

$$EA = \underbrace{\begin{pmatrix} \; \end{pmatrix}}_{2\times 1}\underbrace{\begin{pmatrix} \; & \; \end{pmatrix}}_{2\times 3}$$

$\neq$

not defined

$$AE = \underbrace{\begin{pmatrix} \; & \; \end{pmatrix}}_{2\times 3}\underbrace{\begin{pmatrix} \; \end{pmatrix}}_{2\times 1}$$

$\neq$

not possible.

Let $A = \begin{pmatrix} 1 & 1 \\ -2 & -1 \\ 1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 & -2 \\ 2 & 1 & -2 \end{pmatrix}$, and $C = \begin{pmatrix} 1 & 1 & -3 \\ -1 & 2 & 0 \\ -3 & -1 & 0 \end{pmatrix}$. Find if possible.

$$AB = \begin{pmatrix} 3 & 0 & -4 \\ -4 & 1 & 6 \\ 5 & 1 & -6 \end{pmatrix}$$

$$BA = \underbrace{\begin{bmatrix} 1 & -1 & -2 \\ 2 & 1 & -2 \end{bmatrix}}_{2\times 3} \underbrace{\begin{bmatrix} 1 & 1 \\ -2 & -1 \\ 1 & 2 \end{bmatrix}}_{3\times 2} = \begin{bmatrix} 1+2-2 & 1+1-4 \\ 2-2-2 & 2-1-4 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -2 \\ -2 & -3 \end{bmatrix}}_{2\times 2}$$

$=$

$$AC = \underbrace{[\quad]}_{3\times 2} \underbrace{[\quad]}_{3\times 3}$$

$\neq$

not possible

$$CA = \begin{pmatrix} -4 & -6 \\ -5 & -3 \\ -1 & -2 \end{pmatrix}$$

$$BC = \begin{pmatrix} 8 & 1 & -3 \\ 7 & 6 & -6 \end{pmatrix}$$

$$CB = \underbrace{(\quad)}_{3\times 3} \underbrace{(\quad)}_{2\times 3}$$

$\neq$

not possible because # col's of C is not equal to the # rows of B.

Practice midterm Fall 2012.

Question 1 (5 points) Consider the following matrices:

$$C_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad C_2 = \begin{bmatrix} -2 \\ -4 \\ -2 \end{bmatrix} \quad C_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad B = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

Determine whether the matrix B can be written as a linear combination of the matrices C_1, C_2, C_3, and if so give a $\boxed{\text{specific}}$ linear combination using the matrix names above.

The problem is asking does there exist x, y, z such that

$$x C_1 + y C_2 + z C_3 = B$$

that is,

$$x \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} -2 \\ -4 \\ -2 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

i.e.

$$\begin{bmatrix} x \\ 2x \\ x \end{bmatrix} + \begin{bmatrix} -2y \\ -4y \\ -2y \end{bmatrix} + \begin{bmatrix} 0 \\ z \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x - 2y \\ 2x - 4y + z \\ x - 2y + z \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} x - 2y = -1 \\ 2x - 4y + z = 1 \\ x - 2y + z = 2 \end{bmatrix}$$

i.e. Does this system have a solution? If so, find one sol'n.

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & -1 \\ 2 & -4 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{array}\right] \xrightarrow{\text{rref}} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left[\begin{array}{ccc|c} 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

$y = s$

$z = 3$

$x - 2y = -1$

$x - 2s = -1$

$x = -1 + 2s$

General solution: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1+2s \\ s \\ 3 \end{bmatrix}$

e.g. $s=0 \rightsquigarrow \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}$ a specific solution

check if it works:

$$\begin{array}{cccccccc} & C_1 & & C_2 & & C_3 & & B \\ -1 & \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} & +0 & \begin{bmatrix} -2 \\ -4 \\ -2 \end{bmatrix} & +3 & \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} & \stackrel{?}{=} & \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \end{array}$$

$$\begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \checkmark$$